# Effective Reinforcement Learning for Remote Sensing Vision-Language Models via Supervised Knowledge Injection

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***Abstract*—While vision-language models (VLMs) have demonstrated impressive capabilities in general-domain visual understanding, their performance in remote sensing (RS) tasks remains limited due to domain discrepancies and the scarcity of RS data in typical vision-language pretraining corpora. Recent advances show that reinforcement learning (RL) can enhance VLMs' reasoning capabilities by guiding models to perform intermediate "thought" steps before answering. However, applying RL directly to RS scenarios is often ineffective, as VLMs lack fundamental domain knowledge to begin with. To address this, we propose a novel paradigm that first equips VLMs with RS knowledge in the RS visual grounding task through supervised fine-tuning (SFT) on a curated dataset of chain-of-thought (CoT) samples specific to RS. We then conduct RL to further strengthen the learned capabilities. Experiments on the in-domain dataset show that our RL-enhanced model outperforms its SFT-only counterpart. Furthermore, evaluations on the out-of-domain dataset demonstrate that our method fosters better generalization in RS reasoning. Our approach offers a promising direction for adapting general-purpose VLMs to domain-specific visual understanding tasks.**

***Index Terms*—Computer vision, deep learning, object detection, reinforcement learning, remote sensing.**

## I. Introduction

VISION-LANGUAGE models (VLMs) [1] have emerged as a powerful paradigm in artificial intelligence, achieving remarkable success in a variety of general domain tasks such as image captioning, visual question answering (VQA), and visual grounding [2], [3], [4], [5]. These models typically frame tasks generatively, aligning linguistic and visual information to tackle complex problems. The proficiency of VLMs has spurred significant interest in their application to specialized domains, notably remote sensing (RS), where the proliferation of Earth observation data necessitates advanced automated interpretation techniques.

Despite their potential, adapting general-purpose VLMs to the specific demands of remote sensing visual grounding (RSVG) [6], [7] presents substantial challenges. A primary hurdle is the domain discrepancy: VLMs are often pretrained on vast datasets of natural, ground-level imagery and associated text. RS imagery, however, possesses unique characteristics, including overhead perspectives, extreme scale variations between objects, highly cluttered backgrounds, diverse and often fine-grained object categories, and distinct spectral properties, especially in non-RGB modalities [8]. This mismatch implies that the visual features and conceptual understanding developed by VLMs during pretraining are often ill-suited for the RS domain.

Consequently, these models typically exhibit a significant knowledge gap regarding RS-specific objects, their contextual relationships, and the specialized language used to describe them in grounding queries [9]. For RSVG, which requires precise localization of a target object within an RS image based on a natural language description, this lack of domain-specific understanding is a critical bottleneck. Furthermore, while RS datasets are becoming more prevalent, those explicitly annotated with detailed, step-by-step reasoning for the RSVG task are scarce. This scarcity impedes the development of models that can not only perform the task but also exhibit interpretable grounding processes. Preliminary evaluations underscore this issue: prominent off-the-shelf VLMs, including MiniGPT-v2 [10], LLaVA-1.5 [11], Qwen-VL-Chat [12], Sphinx [13], and Qwen2.5-VL-3B [14], demonstrate markedly poor performance on the RSVG [6] benchmark, achieving AP@IoU = 0.5 scores below 5. This empirically confirms a profound deficiency in their inherent RS knowledge and grounding capabilities.

Reinforcement learning (RL) has emerged as a powerful paradigm for augmenting the reasoning capabilities of large language models (LLMs) and VLMs [15]. Among these techniques, group relative policy optimization (GRPO) [16] stands out for its innovative approach to policy optimization. By comparing groups of sampled responses, GRPO refines a model's decision-making mechanism, effectively guiding it toward more optimal outputs. A key strength of GRPO lies in its ability to encourage models to generate intermediate "thought" steps, thereby fostering more explicit and robust rea-

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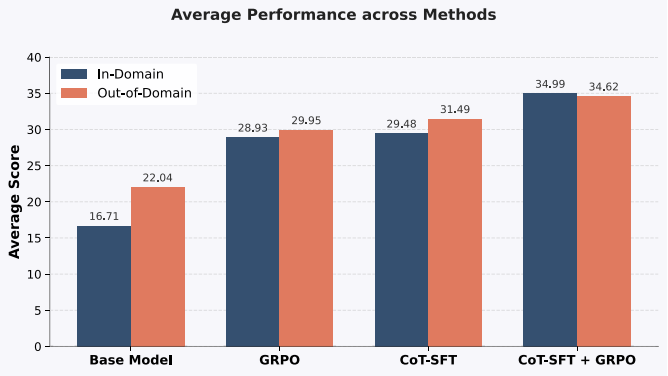



Fig. 1. Performance comparison of four training approaches (base model, GRPO, CoT SFT, and CoT SFT + GRPO) on in-domain and out-of-domain benchmarks for RSVG.

soning processes. This characteristic makes GRPO particularly well-suited for RSVG tasks, which often demand intricate spatial reasoning and a nuanced understanding of textual queries. For example, interpreting instructions like "the small vehicle farthest to the top" from the VRSBench dataset [17] requires a deep comprehension of both visual and linguistic cues. Given these challenges, applying GRPO presents a promising avenue for enhancing the performance of VLMs in the RSVG domain, potentially enabling more accurate and contextually aware visual grounding. However, the direct application of GRPO to standard VLMs for RSVG yields suboptimal results. Fig. 1 provides a comparison of different training strategies. While both GRPO-only and chain-of-thought (CoT)-supervised fine-tuning (SFT) improve over the base model, the combination of CoT-SFT and GRPO consistently performs best across both in-domain and out-of-domain settings. This result suggests that effective adaptation to RSVG requires not only reasoning enhancement, but also prior injection of domain-specific knowledge. GRPO primarily activates and refines latent reasoning abilities, but its effectiveness hinges on the VLM possessing foundational knowledge of the specific domain, including RS objects, terminology, and typical spatial configurations. Without such understanding, GRPO cannot effectively elicit the desired reasoning processes, as evidenced by our experiments where applying GRPO to Qwen2.5-VL-3B on RSVG training data failed to yield significant improvements. This aligns with the central hypothesis of this work: when a VLM lacks requisite RS domain knowledge, techniques like GRPO (or other reasoning enhancement methods) will struggle to drive optimal RSVG performance. Even though GRPO can activate or hone latent reasoning pathways, the model must first recognize RS-specific entities and comprehend their spatial arrangements; otherwise, its "reasoning" is likely to remain superficial or misdirected, underscoring the need for domain-specific foundational understanding before reasoning capabilities can be effectively amplified.

SFT with CoT [18] data has been shown to be an effective strategy for injecting domain knowledge and improving the reasoning capabilities of models as a "cold-start" before applying RL. CoT datasets provide explicit step-by-step reasoning, guiding the model to learn how to approach and solve problems within a specific domain. Unfortunately, existing RSVG datasets [6], [7], [17], [19] typically provide direct query-bounding box pairs without the intermediate reasoning steps crucial for CoT-style SFT.

Existing RSVG-specific methods mainly improve performance through task-specific architectural design, such as multiscale feature fusion, query-guided visual attention (QGVA), and cross-modal alignment. While effective in conventional discriminative grounding settings, these methods are primarily optimized for direct region prediction from image-query pairs, and do not naturally support explicit intermediate reasoning or broader transfer to other multimodal reasoning scenarios. In contrast, our goal is not merely to further optimize a task-specific grounding architecture, but to adapt a general-purpose VLM to the RS domain and endow it with interpretable multistep reasoning ability. This is particularly relevant because many RSVG queries involve compositional spatial constraints, such as relative position, cardinal direction, extreme-location, size comparison, and reference-object-based localization. For such queries, the challenge lies not only in cross-modal alignment, but also in reasoning over multiple linguistic constraints. A generative VLM therefore provides a more natural framework for our objective, as it can jointly model the reasoning process and the final grounding output.

This article proposes a two-stage paradigm: knowledge injection followed by reasoning enhancement. In Stage 1 (knowledge injection via SFT with long CoT), the model is fine-tuned on a new, domain-specific long-CoT SFT dataset to embed RS-specific knowledge and basic grounding procedures into the VLM. Following this, Stage 2 (reasoning enhancement via GRPO) is applied. After acquiring domain knowledge and initial reasoning patterns through SFT, the model undergoes an RL stage using GRPO. This step aims to further refine and strengthen the VLM's capability for multistep reasoning, enabling more accurate and robust visual grounding in RS scenarios.

The main contributions of this work are summarized as follows.

1) *Novel Two-Stage Training Paradigm for VLMs in RSVG:* We introduce a new two-stage training paradigm designed to enhance the reasoning capabilities of VLMs for RSVG. This paradigm uniquely combines long-CoT SFT for comprehensive knowledge injection with GRPO to refine and bolster the model's reasoning pathways.
2) *Methodology for Generating Long-CoT SFT Dataset:* We propose a novel methodology to address the scarcity of reasoning-rich data in the RS domain. This approach leverages a powerful, large-scale "teacher" VLM to generate detailed, step-by-step reasoning chains from existing visual grounding datasets, effectively creating high-quality training data for SFT.
3) *First Long-CoT SFT Dataset for RS Visual Grounding:* By applying our proposed methodology, we have constructed and released the first long-CoT SFT dataset specifically tailored for RSVG. This dataset serves as a valuable new resource for the community,

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enabling future research on advanced reasoning in RS applications.

4) *Demonstrably Improved Reasoning, Interpretability, and Generalization:* We have developed a VLM model that exhibits significantly improved performance on RSVG. Through extensive experiments, our model demonstrates superior reasoning capabilities, enhanced interpretability via explicit thought chains, and robust generalization, validated on both in-domain and out-of-domain benchmarks.

## II. Related Work

### A. Visual Grounding in Remote Sensing

RSVG aims to localize targets in RS images via natural language queries, addressing the challenge of geospatial reasoning in large-scale RS scenes. Early research [6] introduced the RSVG task and proposed GeoVG, a framework that combines language and image encoders with a fusion module to model geographical relationships. The RSVGD dataset [7] later enabled benchmarking of cross-modal methods, where the multilevel cross-modal (MLCM) module [7] leveraged multiscale visual features and text embeddings to handle scale variations.

Subsequent advances focused on enhancing cross-modal fusion and spatial reasoning. VSMR [20] incorporated an adaptive feature selection (AFS) module to suppress background noise and a multistage decoder (MSD) for iterative reasoning on object attributes. LPVA [21] proposed a language-guided progressive attention mechanism to adjust visual features during backbone encoding, while QAMFN [22] integrated QGVA at early visual feature extraction stages to improve text-image alignment. LQVG [23] adopted a language query-based transformer with multiscale cross-modal alignment to retrieve object representations from multiscale visual features, and attribute-prompting networks [24] used learnable prompts to explore object attributes and mitigate confusion between similar targets.

Notably, while these methods advance RSVG, existing datasets primarily provide direct query-to-bbox annotations without explicit step-by-step reasoning paths. This scarcity hinders the development of VLMs that require interpretable grounding processes to enhance generalization ability, as current annotations lack the detailed reasoning cues necessary for VLMs to learn complex cross-modal inference patterns.

### B. Vision-Language Models

The emergence of LLMs has led to transformative progress across a broad spectrum of natural language understanding tasks, laying the foundation for the development of VLMs. Early representative works in this area include [25], [26], [27], which demonstrated the potential of integrating visual and linguistic modalities. Building upon this paradigm, LLaVA [28] leveraged the capabilities of GPT-4 [29] to synthesize large-scale visual instruction data, achieving strong performance in tasks such as visual dialog and multimodal reasoning. This approach has subsequently inspired a series of studies dedicated to constructing and refining visual instruction datasets [11], [30], [31].

Despite these advances, a major constraint in early VLMs was their limited capacity to process high-resolution visual inputs, primarily due to the architectural limitations of their vision encoders [32], [33], [34]. To address this bottleneck, the AnyRes mechanism was introduced [35], [36], [37], enabling dynamic adaptation to images of arbitrary resolutions and aspect ratios. This enhancement significantly broadened the perceptual bandwidth of VLMs and further strengthened their reasoning abilities over complex visual scenes.

Today, several open-source VLM frameworks have become widely adopted within the community, including the LLaVA series [28], [35], [38], QwenVL [14], [39], and InternVL [36], [40], which continue to push the boundaries of multimodal understanding through architectural innovations and data-centric training paradigms.

### C. RL to Strengthen Reasoning Capabilities

The release of OpenAI o1 [41] highlighted the effectiveness of RL in improving the reasoning abilities of LLMs by allowing them to learn directly from outcome-based feedback. More recently, DeepSeek R1 [42] demonstrated that even straightforward rule-based reward mechanisms without relying on dedicated learned reward models [43], [44], [45] can also successfully endow LLMs with sophisticated reasoning skills in an autonomous manner via the GRPO algorithm.

Motivated by the success of RL in enhancing reasoning within text-only LLMs, recent studies have extended the R1 framework to VLMs. Notably, R1-OneVision [46] and R1-V [47] represent pioneering efforts in this direction. R1-OneVision introduced a cross-modal reasoning pipeline that transforms visual inputs into structured representations, which are then used to generate a visual reasoning dataset via a language model. This dataset is first used for supervised pretraining, followed by an RL phase to refine reasoning abilities. Concurrently, R1-V integrated the GRPO algorithm [16] from DeepSeek R1 [42] into VLM training for object counting, remarkably allowing a 3B model to outperform much larger 72B counterparts.

Subsequent work such as VisualThinker-R1-Zero [48] explored applying R1 directly to base VLMs instead of instruction-tuned models, showing even greater gains and the emergence of so-called "visual aha moments." Similarly, MM-Eureka [49] observed such phenomena by applying RLOO [50], [51] to both 8B instruction-tuned and 38B base VLMs. Vision-R1 [52] constructed a multimodal CoT dataset by converting visual input into language sequences to serve as initial training material, followed by GRPO-based RL to further enhance multimodal reasoning. Other studies, such as Curr-ReFT [53], introduced a three-stage RL pipeline with progressively harder reward signals to improve training efficiency, while LMM-R1 [54] proposed a two-phase rule-based RL strategy that first leverages text-only reasoning to bootstrap capabilities, before fine-tuning on more complex multimodal data.

Although most of these efforts target multimodal mathematical reasoning [55], [56], [57], a few (such as Visual-RFT [58] and VLM-R1 [15]) have begun exploring RL in perceptual tasks, aligning more closely with our goals. In contrast, our

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work places a stronger emphasis on enhancing perception capabilities in the context of RS (e.g., RSVG). More importantly, we delve into the key factors that influence the effective adaptation of RL to RS scenarios, providing a promising pathway for tailoring general-purpose VLMs to domain-specific visual understanding tasks.

## III. Methodology

### A. CoT Data Construction

A core objective of our work is to equip the model with robust thinking and reasoning capabilities. In this section, we elaborate on how we construct high-quality CoT data leveraging image-question-answer triples.

To generate CoT-enhanced answers for each instruction data point, we utilize the powerful Qwen2.5-VL-72B model. We employ two distinct methods to produce CoT outputs, subsequently assessing their quality by computing the intersection over union (IoU) between predicted bounding boxes (pred bbox) and the ground-truth bbox (gt bbox) provided in the answers.

*1) Approach 1:* First, we carefully design a prompt to generate both the reasoning process (thinking steps) and the final answer, given only the image and the question. Formally, given an input image $I$, question $Q$, prompt $P_1$, the model produces a CoT response and pred bbox

$$\mathrm{CoT}_1, \mathrm{bbox}_1^{pred} = \mathrm{Qwen}\left(I, Q, \mathrm{P}_1\right). \tag{1}$$

Second, we construct another prompt designed to complete the reasoning process explicitly, providing the model with the image I, question Q, known answer (ground truth) $A^{gt}$, and prompt $P_2$. This initially generated CoT may suffer from significant hallucinations. Thus, we further select it by prompting the model again with the image, question, and generated reasoning steps to regenerate the final answer

$$\mathrm{CoT}_2 = \mathrm{Qwen}\left(I, Q, A^{gt}, \mathrm{P}_2\right) \tag{2}$$

$$\mathrm{bbox}_2^{pred} = \mathrm{Qwen}\left(I, Q, \mathrm{CoT}_2, \mathrm{P}_2\right). \tag{3}$$

**P1: Prompt to generate both the reasoning process (thinking steps) and the final answer, given only the image and the question**

First think about the reasoning process in the mind and then provide the user with the answer. The reasoning process and answer are enclosed within <think> </think> and <answer> </answer> tags, respectively, i.e., <think> reasoning process here <think>, <answer> answer here <answer>.
{Question}
Note that the image is presented in satellite or aerial view. Output the bbox coordinates of detected objects in JSON format within <answer><</answer>. The bbox coordinate format should be:
[{''bbox_2d'': [x1, y1, x2, y2]}]
If no targets are detected in the image, simply respond with "[]".

**P2: Prompt to complete the reasoning process explicitly, providing the model with the image, question, known answer (ground truth)**

You are given an image, a question about the image, and the answer to that question. Your task is to generate a concise but logically sound reasoning process (chain of thought) that explains how to arrive at the answer, based only on the image and the information given in the question. Do not invent details that are not present in the image or question.
Format your output as follows:
<think> Step-by-step reasoning or thought process here, grounded in what is actually visible in the image and described in the question. </think> Be factual, concise, and avoid excessive speculation or hallucination. Only refer to features that are clearly observable or explicitly stated.
{Question}
{Answer}

We calculate the IoUs of the pred bbox from these two CoT methods against the ground-truth bounding box ($\mathrm{bbox}^{gt}$, extracted from $A^{gt}$), selecting the CoT with the higher IoU

$$\mathrm{IoU}_i\left(\mathrm{bbox}_i^{pred}, \mathrm{bbox}^{gt}\right) = \frac{\mathrm{bbox}_i^{pred} \cap \mathrm{bbox}^{gt}}{\mathrm{bbox}_i^{pred} \cup \mathrm{bbox}^{gt}} \tag{4}$$

$$\mathrm{CoT}_{final} = \begin{cases} \mathrm{CoT}_1, & \text{if } \mathrm{IoU}_1 \geqslant \mathrm{IoU}_2 \\ \mathrm{CoT}_2, & \text{else} \end{cases} \tag{5}$$

and $\mathrm{IoU}_{final}$ = Corresponding IoU of $\mathrm{CoT}_{final}$

$$\mathrm{CoT}_{Quality} = \begin{cases} \mathrm{High}, & \text{if } \mathrm{IoU}_{final} > 0.7 \\ \mathrm{Low}, & \text{else.} \end{cases} \tag{6}$$

Prompts $\mathrm{P}_1$ and $\mathrm{P}_2$ are listed below.

We clarify that the use of the ground-truth answer in (2) is only for CoT data construction, rather than for final model evaluation or inference. The purpose of this design is to obtain sufficiently accurate and semantically consistent reasoning traces, and to avoid retaining CoT samples that are built around an incorrect answer. Importantly, the generated $\mathrm{CoT}_2$ does not directly contain the final bounding-box coordinates, and thus does not provide explicit coordinate-level leakage. Moreover, the generated CoT is not provided to the final model at test time. During inference, the model receives only the image and the query, and any reasoning process is generated by the model itself.

*2) Approach 2:* In addition, we introduce a more stringent selection criterion to acquire higher quality CoT data. For each instruction data instance, we first generate a CoT-enhanced pred bbox ($\mathrm{bbox}_{cot}^{pred}$) following the method described above. We emphasize that the use of prompt $P_{cot}$ can produce CoT. Next, we produce another prediction using a regular prompt ($P_{no_cot}$) that cannot produce CoT, and the generated bounding box is $\mathrm{bbox}_{no_cot}^{pred}$. We then compute their respective IoUs with the ground truth bounding box, denoted as $\mathrm{IoU}_{cot}$ and $\mathrm{IoU}_{no_cot}$

$$\mathrm{bbox}_{cot}^{pred} = \mathrm{Qwen}\left(I, Q, P_{cot}\right) \tag{7}$$

$$\mathrm{bbox}_{no_cot}^{pred} = \mathrm{Qwen}\left(I, Q, P_{no_cot}\right) \tag{8}$$

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$$\mathrm{IoU}_{cot} = \mathrm{IoU}\left(\mathrm{bbox}_{cot}^{pred}, \mathrm{bbox}^{gt}\right) \tag{9}$$

$$\mathrm{IoU}_{no_cot} = \mathrm{IoU}\left(\mathrm{bbox}_{no_cot}^{pred}, \mathrm{bbox}^{gt}\right). \tag{10}$$

Let $\Delta\mathrm{IoU} = \mathrm{IoU}_{cot} - \mathrm{IoU}_{no_cot}$. We classify CoT data as high-quality when the following conditions are satisfied:

$$\mathrm{CoT}_{Quality} = \begin{cases} \mathrm{High} & \text{if} \quad \Delta\mathrm{IoU} > 0.3 \text{ and } \mathrm{IoU}_{cot} > 0.5 \\ \mathrm{Low} & \text{else.} \end{cases} \tag{11}$$

The rationale behind this is to select CoT data that truly aids the model in grounding the target object. Thus, we set a margin of 0.3 and further restrict the final results to ensure that the grounding result is meaningful (i.e., IoU > 0.5).

These filtering processes help ensure that the resulting CoT data significantly enhances the model's reasoning accuracy, capturing cases where reasoning steps transform initially incorrect or imprecise predictions into correct and precise ones.

### B. Supervised Fine-Tuning

*1) Model Architecture:* The typical model architecture of VLMs consists of a vision encoder $\mathcal{F}_{\mathrm{v}}$, a vision projector layer $W_{\mathrm{v}}$, and an underlying LLM $\mathcal{F}_{\mathrm{llm}}$. Generally, given an input image *Img*, the visual encoder $\mathcal{F}_{\mathrm{v}}$ first extracts visual features $Z_{\mathrm{v}}$. These features are then projected into the word embedding space of the LLM via the linear layer $W_{\mathrm{v}}$, producing a sequence of visual tokens. Concurrently, the word embedding matrix of the LLM is adopted to transform the user's instruction *Ins* into a set of textual tokens $X_{\mathrm{t}}$. The resulting visual tokens and textual tokens are subsequently concatenated collaboratively to form the final input $X_{\mathrm{input}}$ to the LLM. This process can be formalized as

$$Z_{\mathrm{v}} = \mathcal{F}_{\mathrm{v}}(Img) \tag{12}$$

$$X_{\mathrm{v}} = W_{\mathrm{v}}(Z_{\mathrm{v}}) \tag{13}$$

$$X_{\mathrm{t}} = W_{\mathrm{t}}(Ins) \tag{14}$$

$$X_{\mathrm{input}} = [X_{\mathrm{v}},\ X_{\mathrm{t}}]. \tag{15}$$

*2) Training:* For SFT of the VLM, we adopt an autoregressive loss that is computed exclusively over the target answer tokens. Given an input sequence $X_{\mathrm{input}}$, the model is trained to maximize the likelihood of the target answer sequence $X_{\mathrm{a}} = \{x_1, x_2, \ldots, x_L\}$. The conditional probability of generating the answer sequence is formulated as

$$p\left(X_{\mathrm{a}}|X_{input}\right) = \prod_{i}^{L} p_\theta\left(x_i|X_{\mathrm{input}}, x_1, \ldots, x_{i-1}\right) \tag{16}$$

where $\theta$ are the trainable parameters, and $x_1, \ldots, x_{i-1}$ are the answer tokens before the current prediction token $x_i$. In our implementation, we train all parameters of the VLM, that is $\theta = \{\mathcal{F}_{\mathrm{v}}, W_{\mathrm{v}}, \mathcal{F}_{\mathrm{llm}}\}$.

### C. Group Relative Policy Optimization

*1) Abstraction of GRPO:* In contrast to traditional RL methods such as PPO [59], which rely on an additional critic network to estimate the value of the current policy, GRPO bypasses the need for a separate critic by directly performing comparisons among multiple candidate responses. Given a prompt q, GRPO samples a set of N candidate outputs $\{o_1, o_2, \ldots, o_N\}$ from the policy $\pi_\theta$, and each response $o_i$ is assessed using a reward function $R(q, o_i)$, which quantifies the quality of the output conditioned on the input query. To compute relative advantages among the responses, GRPO standardizes the raw rewards by calculating their mean and standard deviation, and then derives the advantage for each candidate accordingly

$$A_i = \frac{r_i - \mathrm{mean}\{r_1, r_2, \ldots, r_N\}}{\mathrm{std}\{r_1, r_2, \ldots, r_N\}} \tag{17}$$

where $A_i$ denotes the normalized advantage of the candidate response $o_i$ relative to other samples within the group. GRPO optimizes the policy $\pi_\theta$ by encouraging responses with higher relative advantages, using the following objective:

$$\mathcal{J}_{\mathrm{GRPO}}(\theta) = \mathbb{E}_{\{o_i\}_{i=1}^{N} \sim \pi_{\theta_{\mathrm{old}}}} \left[ \frac{1}{N} \sum_{i=1}^{N} \left( \min\left[r_i A_i,\ \bar{r}_i A_i\right] - \beta D_i \right) \right] \tag{18}$$

where

$$r_i = \frac{\pi_\theta(o_i|q)}{\pi_{\theta_{\mathrm{old}}}(o_i|q)}, \quad \bar{r}_i = \mathrm{clip}\left(r_i,\ 1-\epsilon,\ 1+\epsilon\right) \tag{19}$$

and

$$D_i = \mathbb{D}_{\mathrm{KL}}\left[\pi_\theta(\cdot|q) \,\|\, \pi_{\mathrm{ref}}(\cdot|q)\right]. \tag{20}$$

Here, $D_i$ denotes the Kullback–Leibler divergence between the current policy $\pi_\theta$ and the reference policy $\pi_{\mathrm{ref}}$, which serves as a regularization term to prevent large deviations. The clipping mechanism in $\bar{r}_i$ stabilizes training by constraining the policy update ratio.

The reward affects training through the GRPO objective rather than serving only as an evaluation signal. After each sampled response is assigned a reward, the reward is normalized into a relative advantage according to (17). This advantage is then inserted into the objective in (18), and the policy parameters are updated by gradient-based optimization of this objective. As a result, responses with higher relative advantage contribute gradients that increase their likelihood under the updated policy, whereas responses with lower relative advantage are suppressed. Therefore, the reward design directly determines which kinds of reasoning-and-answer trajectories are reinforced during optimization.

*2) Reward Design:* To effectively guide the VLM toward producing both semantically correct and structurally compliant outputs, we incorporate a dual-reward mechanism consisting of an accuracy reward and a format reward. This design follows the default reward setting of VLM-R1 [15], which targets a closely related visual grounding task. By combining these two reward types, we aim to ensure that the model not only generates accurate answers but also adheres to the expected response structure required for downstream applications such as automated evaluation and deployment.

*Accuracy Reward:* RSVG is a task that requires the model to localize an object within an image based on a natural language referring expression. Formally, let q be the input question, $b^*$ the ground-truth bounding box, o the textual output generated by the VLM, and f a postprocessing function that parses the

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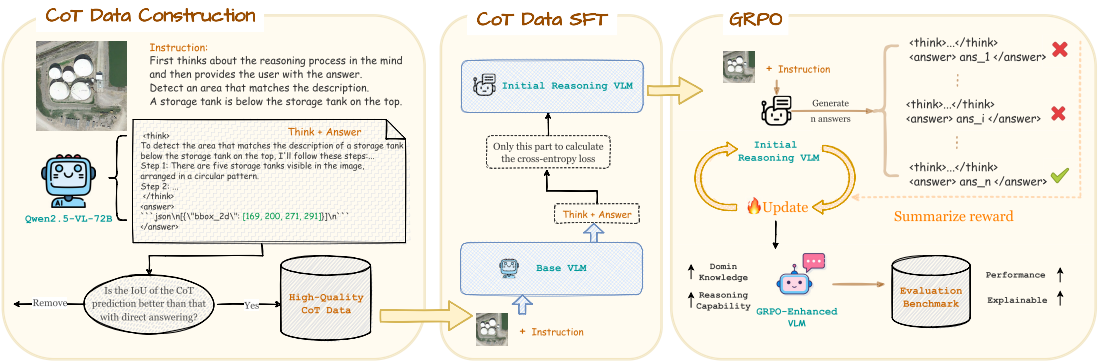



Fig. 2. Schematic illustration of the proposed two-stage training paradigm.

pred bbox from o. The accuracy reward is computed using the IoU between the predicted and ground-truth boxes

$$R_{acc}(q,o) = \mathrm{IoU}(b^*, f(o)). \tag{21}$$

This reward encourages the model to generate outputs that not only reference a box but do so in a spatially precise manner. A higher IoU indicates better alignment between the prediction and the reference, thus reflecting stronger perceptual grounding.

Unlike cross-entropy-based supervision, which treats box coordinates as discrete output tokens, the IoU reward provides a geometry-aware outcome signal. As a result, a predicted box that is not exactly identical to the ground-truth coordinates but is still spatially close can receive a relatively high reward, making the optimization better aligned with the actual grounding objective.

*Format Reward:* In addition to correctness, it is important that the output follows a well-defined format to facilitate reliable parsing and interpretation. We therefore define a binary format reward that checks whether the model's output conforms to a structured response pattern. Specifically, the response must be enclosed in tags and include a JSON-style bounding box of the form `[x1, y1, x2, y2]` nested within reasoning tags:

`(<think>...</think><answer>{...[x1, y1, x2, y2]...}</answer>)`

The format reward is defined as

$$R_{\text{format}}(q,o) = \begin{cases} 1, & \text{if output follows the expected format} \\ 0, & \text{otherwise.} \end{cases} \tag{22}$$

This component acts as a structural regularizer, helping the model maintain output consistency and reducing the burden of downstream postprocessing or error handling.

The format reward mainly constrains the response structure, encouraging the model to preserve explicit reasoning traces and a parsable answer format during RL training.

By jointly optimizing both rewards, the training process is better aligned with the end-task objectives, improving both the functional correctness and robustness of the model's predictions in RS visual grounding scenarios.

### D. Two-Stage Training Paradigm

As shown in Fig. 2, our training framework adopts a two-stage paradigm to inject the base model with domain-specific knowledge and robust reasoning capabilities.

In the first stage, we leverage high-quality reasoning data that includes explicit CoT annotations introduced in Section III-A. This curated dataset enables the model to acquire not only fundamental domain knowledge but also structured reasoning processes. By SFT on these CoT-augmented examples, the base model learns to generate detailed step-by-step rationales when presented with new image-question pairs, thus enhancing its interpretability and bringing the reasoning ability.

The second stage applies GRPO to further boost the model's reasoning ability. In this phase, the model is optimized with RL, using the reasoning format and final prediction accuracy (IoU between bounding boxes) as a reward signal. The GRPO procedure encourages the model to refine its reasoning strategies and maximize the expected quality of generated answers. This reinforcement step enhances the initial reasoning ability injected during the SFT, resulting in a model that demonstrates stronger and more reliable reasoning on visual grounding tasks.

## IV. Experiments and Results

### A. Datasets

Our experimental setup leverages a combination of existing and newly generated datasets to effectively train and evaluate our model's capabilities in the RS visual grounding task.

*Training Dataset:* The foundation for our training data is the visual grounding subset from the Falcon_SFT instruction-tuning dataset [19], which consists of 32 496 samples. From this initial pool, we curate specific subsets for our two training stages as follows.

1) *SFT Training Data:* For the SFT phase, we prioritize high-quality reasoning data. We first generate long-CoT annotations for all 32 496 samples. Then, to assess the quality of each CoT sample, we measure the IoU between the bounding box generated within the CoT and the ground-truth bounding box. We select around 7000 samples using high $CoT_{Quality}$ CoT Data based on the mix strategy of approach 1 and approach 2 mentioned

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in Section III-A to form our high-quality SFT training set.

2) *GRPO Training Data:* For the RL phase, we aim to build a curriculum that covers a wide range of task complexities. We use the IoU between a generated bounding box and the ground-truth bounding box as a proxy for task difficulty. We then construct a diverse 10 000-sample training set by uniformly sampling 1000 data points from each 0.1 IoU interval between 0.0 and 1.0 (i.e., b[0.1, 0.2), . . . , [0.9, 1.0]). This strategy ensures the model is exposed to a balanced mix of easy, medium, and hard examples during GRPO training.

*Evaluation Datasets:* To rigorously assess the performance and generalization of our model, we employ both in-domain and out-of-domain evaluation benchmarks.

1) *In-Domain Evaluation:*
   a) *DIOR-RSVG:* Derived from the DIOR object detection dataset, DIOR-RSVG is characterized by significant object size variations, diverse image quality, and class diversity. We evaluate our models on the combination of its validation set (3829 samples) and test set (7500 samples).
   b) *RSVG:* For our in-domain evaluation, we use the combined validation and test sets of the RSVG dataset, totaling 2428 samples across 1276 images. The images in this dataset are large-scale with varied spatial resolutions, captured from multiple sensors and platforms like Google Earth. A key characteristic of the RSVG dataset is its use of complex spatial relationship descriptions that differ from those in natural images. Queries often employ cardinal directions (e.g., "east side") and numerical distances, posing a significant reasoning challenge that contemporary models often fail to address adequately. This makes it a robust benchmark for evaluating in-domain performance on nuanced reasoning.
2) *Out-of-Domain Evaluation:*
   a) *VRSBench:* To assess the model's zero-shot generalization and reasoning capabilities in more challenging, unseen scenarios, we use the VRS-Bench test set. VRSBench is a comprehensive benchmark comprising 29 614 images, 52 472 object references, and 123 221 question-answer pairs, with a test set of 9350 images. It is specifically designed to test VLMs with complex queries that require nuanced spatial reasoning and discrimination between multiple similar objects. This makes it an ideal testbed for evaluating the zero-shot generalization of the reasoning abilities cultivated by our SFT + GRPO training paradigm.

### B. Statistics for CoT Data

To better characterize the CoT SFT dataset for RS visual grounding, we summarize its basic statistics from three aspects: CoT length distribution, vocabulary characteristics, and spatial relation types.

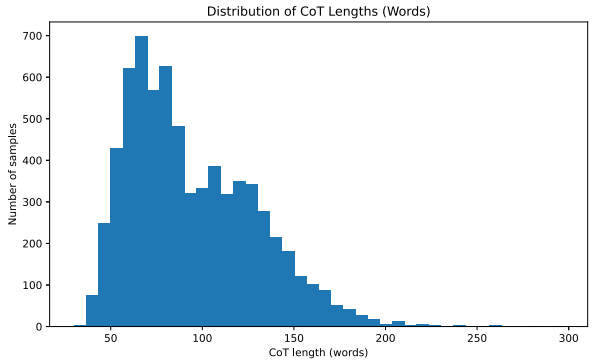



Fig. 3. Distribution of CoT lengths in the CoT SFT dataset, measured in words.

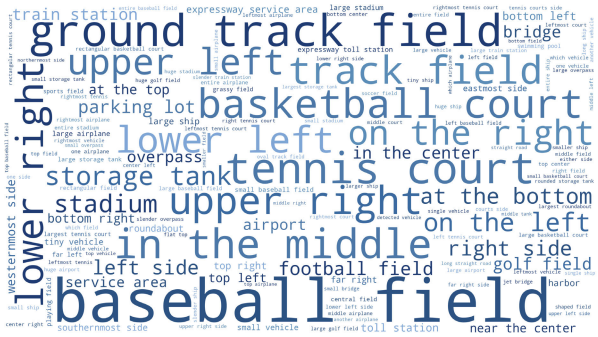



Fig. 4. Word cloud of representative content-bearing terms in the CoT SFT dataset.

The final SFT dataset contains 6974 high-quality CoT samples. All samples contain valid <think> fields and valid bounding-box answers, indicating that the final curated set is structurally complete and well-formed. Fig. 3 shows the distribution of CoT lengths measured in words. Overall, the distribution is right-skewed: the average CoT length is 94.7 words, the median length is 87 words, and the interquartile range is 67–119 words. Most samples therefore contain moderately long reasoning traces, while a smaller number of instances include substantially longer reasoning chains, with the maximum length reaching 297 words. This suggests that the dataset covers both concise and relatively elaborate reasoning styles.

In terms of vocabulary, the dataset contains 2554 unique words. We focus on representative words and semantic phrases. As illustrated in Fig. 4, the vocabulary is strongly grounded in both RS object categories and spatial expressions. Frequent terms and phrases are mainly associated with sports facilities, transportation objects, built structures, and spatial descriptions, such as tennis court, baseball field, ground track field, storage tank, vehicle, airplane, ship, as well as in the middle, on the right, upper left, and at the bottom. This suggests that the generated CoTs are not merely generic reasoning templates, but are closely grounded in the semantic categories, object attributes, and spatial relations that characterize RS scenes.

We also analyze the spatial relation types appearing in the referring expressions. As summarized in Table I, the

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TABLE I
DISTRIBUTION OF SPATIAL RELATION TYPES IN THE RELEASED LONG-COT SFT DATASET

| Relation type | Count | Example |
|---|---|---|
| Relative position | 5212 | *in the middle* |
| Size comparison | 1681 | *a tiny vehicle on the lower left* |
| None | 699 | *the gray train station* |
| Extreme reference | 148 | *the southernmost side* |
| Cardinal direction | 95 | *northwest of the tennis court* |
| Distance | 15 | *close to a parking lot* |
| Containment | 9 | *within 10 meters of the tennis court* |
| Count | 5 | *four tennis courts* |

dataset is dominated by relative position relations, with 5212 occurrences, followed by size comparison relations (1681 appearances). In addition, a smaller but non-negligible proportion of samples involve more challenging relation types, including extreme references (148) and cardinal directions (95). These results show that the dataset is not limited to simple location descriptions such as "left" or "right," but also includes more complex forms of spatial reasoning. Since a single query may contain multiple relation types, these categories are multilabel and therefore not mutually exclusive.

Note that these categories are multilabel rather than mutually exclusive, since a single referring expression may contain more than one spatial relation type.

Overall, these statistics indicate that the CoT SFT dataset is diverse in reasoning length, semantically grounded in RS object categories, and structurally rich in spatial relation patterns, making it a suitable resource for supervised knowledge injection in RSVG.

### C. Evaluation Metrics

To quantify the performance of our models on the visual grounding task, we employ distinct metrics tailored to the characteristics of each evaluation dataset.

*1) In-Domain Metrics:* For the DIOR-RSVG and RSVG datasets, we adopt the average precision at IoU = 0.5 (AP@50) metric, consistent with the evaluation protocol outlined in [19]. The AP@50 metric evaluates the average precision of pred bbox where the IoU with the ground-truth boxes meets or exceeds 50%. This metric balances precision and recall across various confidence thresholds. It is defined as

$$\text{AP@IoU=50} = \frac{1}{|D|}\sum_{d\in D}\text{Precision}(d)\times\text{Recall}(d). \quad (23)$$

Here, $D$ represents the set of all detections, and the Precision and Recall values are computed at an IoU threshold of 0.5.

*2) Out-of-Domain Metrics:* For the VRSBench dataset, we utilize the accuracy at IoU = 0.5 (Acc@0.5) metric, as outlined in the benchmark's evaluation guidelines. A prediction is deemed correct if the IoU between the pred bbox ($B_{\text{pred}}$) and the ground-truth bounding box ($B_{\text{gt}}$) is at least 0.5. The overall accuracy is calculated as the ratio of correct predictions to the total number of samples. The formula is

$$\text{Acc@0.5} = \frac{1}{N}\sum_{i=1}^{N} I\left(\text{IoU}\left(B_{\text{pred}_i}, B_{\text{gt}_i}\right)\geq 0.5\right) \quad (24)$$

TABLE II
COMPARISON OF MODELS ON IN-DOMAIN DATASETS. MINIGPT-V2, LLAVA-1.5, QWEN-VL-CHAT, AND SPHINX ARE GENERAL-PURPOSE VLMS. GEOCHAT AND LHRS-BOT ARE VLMS SPECIFICALLY TRAINED ON RS-RELATED DATASETS. QWEN2.5-VL-3B SERVES AS THE PRETRAINED BASELINE MODEL FOR OUR EXPERIMENTS. COT-SFT+GRPO IS AN ENHANCED MODEL BUILT UPON THE COT-SFT MODEL

| Models | RSVG AP@IoU=0.5 | DIOR-RSVG AP@IoU=0.5 |
|---|---|---|
| MiniGPT-v2 | 1.77 | 28.89 |
| LLaVA-1.5 | 0.17 | 12.09 |
| Qwen-VL-Chat | 3.63 | 31.53 |
| Sphinx | 0.00 | 0.94 |
| GeoChat | 0.74 | 21.02 |
| LHRS-Bot | 1.32 | 11.83 |
| Qwen2.5-VL-3B (Baseline) | 4.96 | 28.46 |
| GRPO-Only | 10.93 | 46.93 |
| CoT-SFT | 9.06 | 49.91 |
| CoT-SFT+GRPO | **13.10** | **56.88** |

where $N$ is the total number of test samples, and $I(\cdot)$ is the indicator function, returning 1 if the condition is met and 0 otherwise. This metric assesses the model's capability to accurately localize the referred object based on textual descriptions.

### D. Experiments Setting

All experiments are conducted using the pretrained Qwen2.5-VL-3B model as the foundational VLM. Training for all models was performed on a high-performance computing cluster equipped with 8 NVIDIA H100 80G GPUs.

For the SFT phase, where we fine-tune the model on our generated long-CoT dataset and regular SFT dataset, we use the AdamW optimizer. The learning rate is set to $1e^{-5}$ with a total batch size of 32. We employ a cosine learning rate scheduler with a warm-up ratio of 0.1 to ensure stable convergence.

In the RL phase, the GRPO algorithm is configured with a learning rate of $1e^{-6}$. During training, the policy model generates N = 16 candidate responses for each input with a generation temperature of 0.9 to encourage exploration. The KL divergence regularization coefficient, $\beta$, is set to 0.04.

### E. Comparison Experiments

As shown in Table II, general-purpose VLMs such as MiniGPT-v2, LLaVA-1.5, and Qwen-VL-Chat achieve low AP@IoU = 0.5 scores on both RSVG and DIOR-RSVG, indicating a lack of domain knowledge for RS visual grounding tasks. Similarly, the baseline (Qwen2.5-VL-3B) shows suboptimal performance.

Models trained with only GRPO demonstrate moderate improvements, and the model cannot perform reasoning (only output bounding box coordinates). Moreover, the lack of explicit domain knowledge still limits its overall effectiveness. Introducing CoT-based SFT (CoT-SFT) further boosts performance and enables the model to generate detailed reasoning chains.

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TABLE III

PERFORMANCE COMPARISON OF MODELS ON THE OUT-OF-DOMAIN DATASET. MODELS MARKED WITH * HAVE BEEN FINE-TUNED ON THE VRSBENCH TRAINING SET, WHILE OTHERS ARE EVALUATED IN A ZERO-SHOT SETTING

| Models | VRSBench Acc@0.5 |
|---|---|
| MiniGPT-v2* | 35.8 |
| LLaVA-1.5* | 36.3 |
| GeoChat* | **39.6** |
| Qwen2.5-VL-3B (Baseline) | 22.04 |
| GRPO-Only | 29.95 |
| CoT-SFT | 31.49 |
| CoT-SFT+GRPO | 34.62 |

The best results are achieved by combining CoT-SFT with GRPO (CoT-SFT+GRPO), which significantly enhances both reasoning ability and generalization, yielding the highest AP@IoU = 0.5 scores across both datasets. This confirms that SFT with CoT data and RL complement each other, providing substantial gains for domain-specific visual grounding. Simply relying on pretrained or generalist models is insufficient for RS visual grounding. Incorporating high-quality CoT data and RL is crucial for boosting both performance and interpretability.

Table III summarizes model performance on the out-of-domain (OOD) VRSBench dataset. Models fine-tuned on VRSBench (MiniGPT-v2, LLaVA-1.5, and GeoChat) achieve the highest Acc@0.5 scores, with GeoChat performing best at 39.6. In contrast, the baseline Qwen2.5-VL-3B and all enhanced variants are evaluated in a zero-shot setting, reflecting a more realistic OOD generalization scenario.

Among the zero-shot models, both GRPO-Only and CoT-SFT improve substantially over the baseline (29.95 and 31.49 versus 22.04), demonstrating their effectiveness in enhancing model robustness and transferability. The combined CoT-SFT+GRPO model further improves performance to 34.62, approaching the scores of the in-domain fine-tuned models.

These results indicate that although there is still a gap compared to in-domain fine-tuned models, the integration of SFT with reasoning chain (CoT-SFT) and RL (GRPO) notably enhances the generalization capability of the model on OOD tasks. This highlights the value of structured reasoning and reinforcement-driven adaptation for building more robust and transferable VLMs in the RS visual grounding task.

### F. Ablation Study

*1) Effect of Data Complexity in GRPO Training:* We investigate the impact of the training data's difficulty distribution on the efficacy of the GRPO stage. To do this, we curate three distinct 10-k-sample training datasets based on task complexity. The complexity of each (image, query) pair is proxied by the IoU score of the corresponding CoT-generated bounding box from our data generation phase (Section II-A). A high IoU indicates an "Easy" instance, while a low IoU signifies a "Hard" one. This ablation is conducted on the model that was trained with 7K long-CoT SFT data. The three sampling strategies are as follows.

TABLE IV

COMPARISON OF AVERAGE SCORES BASED ON DATA COMPLEXITY. "UNIFORM SAMPLING" REFERS TO THE METHOD OF EVENLY SAMPLING DATA POINTS ACROSS IOU INTERVALS TO ENSURE A BALANCED MIX OF EASY, MEDIUM, AND HARD EXAMPLES FOR GRPO DATA

| Data Complexity | Average Score |
|---|---|
| Easy | 33.86 |
| Hard | 33.21 |
| Uniform Sampling | 34.87 |

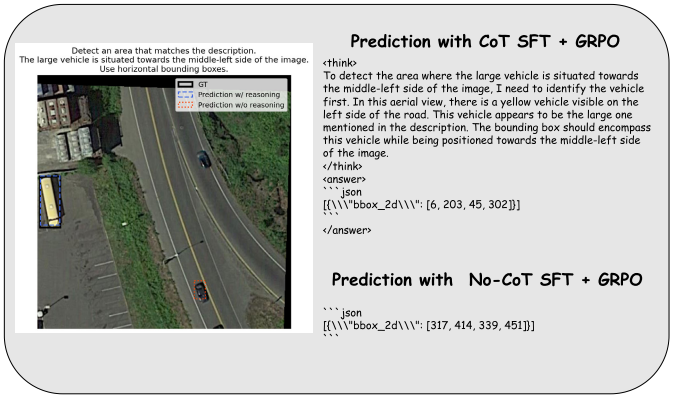



Fig. 5. Example outputs comparison between models trained with CoT SFT + GRPO and No-CoT SFT + GRPO.

1) *Easy:* The 10 000 samples with the highest IoU scores.
2) *Hard:* The 10 000 samples with the lowest IoU scores (where IoU > 0.1).
3) *Uniform Sampling:* Our proposed method, which balances the curriculum by sampling uniformly across the full IoU range.

The results in Table IV show that uniform sampling achieves the highest score (34.87). Training exclusively on easy data (33.86) is less effective, likely because it fails to sufficiently challenge the model to improve its reasoning on more difficult cases. In contrast, training only on hard data yields the lowest score (33.21), suggesting that focusing the RL process solely on the most challenging instances is suboptimal for overall performance. This experiment confirms that a balanced curriculum, exposing the model to a diverse range of problem difficulties, is critical for maximizing the benefits of GRPO.

*2) Comparison on Knowledge Injection Method:* This ablation study assesses the importance of using CoT data during the initial SFT phase. We compare our proposed CoT SFT + GRPO method with a No-CoT SFT + GRPO baseline. In our approach, the model is first fine-tuned on the 7k long-CoT SFT dataset before undergoing GRPO training. The baseline, in contrast, is fine-tuned on the original 7-k data before undergoing the same GRPO stage.

Qualitative results in Fig. 5 show that the CoT SFT + GRPO model generates an explicit intermediate reasoning process before predicting the final bounding box, whereas the No-CoT SFT + GRPO model directly outputs only the bounding box.

This difference is further supported by the training dynamics shown in Figs. 6 and 7. Fig. 6 reports the format reward, which directly measures whether the model produces outputs

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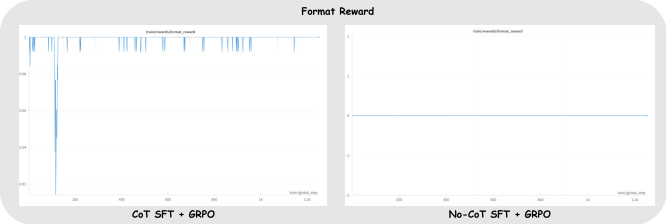



Fig. 6. Training log of format reward of CoT SFT + GRPO and No-CoT SFT + GRPO.

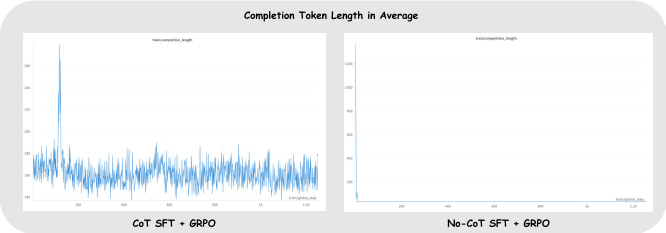



Fig. 7. Training log of completion token length of CoT SFT + GRPO and No-CoT SFT + GRPO.

in the required "think-then-answer" format. Therefore, this metric is not merely a generic training signal, but a direct indicator of whether explicit reasoning-form outputs are successfully elicited during GRPO. As shown in Fig. 6, the CoT SFT + GRPO model maintains nonzero format reward throughout training, indicating that the model can stably generate reasoning traces together with the final answer. In contrast, the No-CoT SFT + GRPO model remains near zero, suggesting that, without CoT-based supervised initialization, GRPO alone fails to establish the desired reasoning structure and mainly optimizes only the final answer format. This result supports our claim that CoT-based SFT provides an essential initialization for reasoning-form behavior, making the model amenable to subsequent RL.

Fig. 7 provides additional evidence from the perspective of output length. The CoT SFT + GRPO model consistently produces substantially longer completions than the No-CoT SFT + GRPO model. Importantly, this increase in length should not be interpreted in isolation as simply generating more tokens. Combined with the nonzero format reward in Fig. 6 and the qualitative examples in Fig. 8, it indicates that the additional tokens correspond to explicit intermediate reasoning steps rather than arbitrary verbosity. By contrast, the short completions of the No-CoT SFT + GRPO model are consistent with direct box prediction without an interpretable reasoning process. Taken together, these training dynamics support our claim that CoT-based SFT does not only inject RS knowledge, but also preserves and activates an explicit reasoning trajectory that GRPO can further refine, ultimately leading to better interpretability and stronger grounding performance.

These findings suggest that while standard SFT (No-CoT) can inject domain-specific knowledge, it may inadvertently cause the model to "unlearn" or suppress the latent reasoning abilities present in the base VLM. Consequently, the subsequent GRPO phase struggles to elicit a reasoning process that is no longer readily accessible. By fine-tuning on CoT data, we not only inject knowledge but also preserve and structure the model's inherent reasoning capabilities. This makes the model more amenable to the GRPO stage, allowing it to more effectively explore and refine its problem-solving strategies, leading to a more powerful and interpretable final model.

TABLE V

COMPARISON OF AVERAGE SCORES FOR DIFFERENT COT DATA GENERATION STRATEGIES. "APPROACH 1" AND "APPROACH 2" ARE INTRODUCED IN SECTION III-A. "MIX" REPRESENTS THE USAGE OF A MIX SET OF BOTH APPROACHES

| CoT Approach | Average Score |
|---|---|
| Approach 1 | 33.79 |
| Approach 2 | 34.31 |
| Mix | 34.87 |

TABLE VI

CROSS-BACKBONE EVALUATION ON DIOR-RSVG (AP@IOU = 0.5)

| Models | Baseline | SFT-Only | GRPO-Only | SFT + GRPO |
|---|---|---|---|---|
| Qwen2.5-VL-3B | 28.46 | 49.91 | 46.93 | **56.88** |
| Qwen3-VL-4B | 49.63 | 59.20 | 45.67 | **62.63** |
| Intern3.5-VL-4B | 57.70 | 63.77 | 64.57 | **68.81** |

*3) CoT Data Generation Strategy:* Table V compares the average scores of different CoT data generation approaches. Both Approach 1 and Approach 2 yield similar performance, with average scores of 33.79 and 34.31, respectively. Notably, the mixed approach—which combines data from both methods—achieves the highest score at 34.87. This suggests that leveraging a diverse set of reasoning sources can further enhance model performance, likely due to improved robustness and generalization when exposed to varied reasoning patterns during training.

*4) Cross-Backbone Generalization:* To evaluate whether the proposed framework is specific to a single VLM backbone, we further conduct additional experiments on two more mainstream VLMs on the DIOR-RSVG dataset, namely Qwen3-VL-4B-Instruct [60] and Intern3.5-VL-4B-Instruct [61]. For a fair comparison, all three backbones, including Qwen2.5-VL-3B, Qwen3-VL-4B, and Intern3.5-VL-4B, are evaluated using the same dataset split, overall training settings, and evaluation protocol. For each backbone, we compare four settings: the baseline model, CoT-SFT only, GRPO only, and the full two-stage framework (CoT-SFT + GRPO). The results are reported in Table VI.

Several observations can be made. First, the proposed framework consistently improves performance across all tested backbones. In particular, the full two-stage setting (CoT-SFT + GRPO) achieves the best result for all three models, and in every case surpasses not only the baseline, but also each single-stage variant considered separately, namely CoT-SFT only and GRPO only. These results suggest that the effectiveness of our framework is not limited to a single pretrained VLM backbone, and further support the complementary roles of supervised knowledge injection and reinforcement-based reasoning refinement. Second, the relative contributions of CoT-SFT and GRPO vary across

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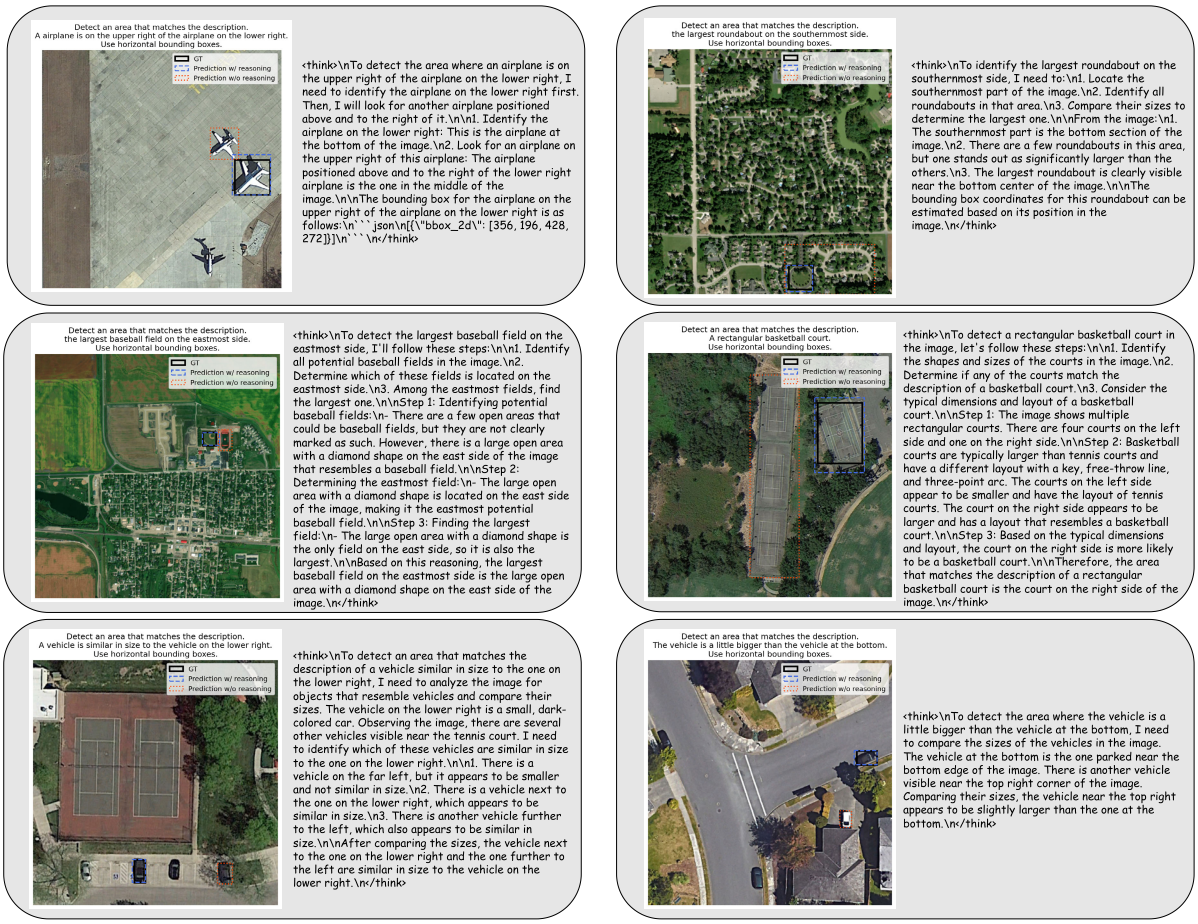



Fig. 8. Some examples of the reasoning process that help the model ground the object.

TABLE VII
SENSITIVITY ANALYSIS OF THE IOU THRESHOLDS USED IN APPROACH 1 AND APPROACH 2

| Approach 1 / Approach 2 | 0.4 | 0.5 | 0.6 |
|---|---|---|---|
| 0.6 | 54.92 | 55.83 | 53.61 |
| 0.7 | 55.68 | **56.88** | 54.72 |
| 0.8 | 51.84 | 50.57 | 48.93 |

backbones. On Qwen3-VL-4B, CoT-SFT yields a larger gain than GRPO-only, whereas on Intern3.5-VL-4B, GRPO-only already achieves competitive improvement over the baseline. We believe this difference reflects variation in the initial remote-sensing knowledge and latent reasoning ability of different pretrained VLMs.

*5) Sensitivity to CoT Filtering Thresholds:* We conducted experiments on the DIOR-RSVG dataset with different IoU threshold settings for Approach 1 and Approach 2 to investigate the most suitable threshold pair for CoT data construction. Specifically, the threshold in Approach 1 was varied among 0.6, 0.7, and 0.8, while the threshold in Approach 2 was varied among 0.4, 0.5, and 0.6.

As shown in Table VII, the setting of 0.7 for Approach 1 and 0.5 for Approach 2 achieves the best performance. This suggests that the optimal threshold choice should balance two competing factors: the quality of retained CoT samples and the coverage of the resulting supervision. If the thresholds are too loose, more noisy or unreliable CoT samples are retained, which dilutes the supervision signal in CoT-SFT and weakens the quality of the injected reasoning patterns. However, if the thresholds are made overly strict, the number of retained samples decreases much more sharply, especially for Approach 1, leading to insufficient RS-related supervision coverage. We found this loss of data quantity and diversity is more harmful than introducing a moderate amount of noise, because CoT-SFT is expected not only to fit the target outputs, but also to inject remote-sensing knowledge and diverse reasoning patterns as a cold start for the subsequent GRPO stage.

## V. REASONING PROCESS DEMONSTRATION

Fig. 8 presents representative examples of the reasoning processes generated by the model. These cases show that the proposed two-stage framework enables the model to perform interpretable multistep reasoning for RS visual grounding, instead of directly producing only the final prediction.

Specifically, the examples illustrate several concrete reasoning patterns. For the airplane example, the model first identifies the airplane on the lower right as the reference object and then locates the target airplane on its upper right. For the roundabout and baseball-field examples, the model first determines the relevant extreme spatial region, such as the

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eastmost side, and then selects the largest candidate within that region. For the vehicle examples, the model explicitly compares object sizes to identify vehicles that are similar to, or slightly larger than, a reference vehicle. These results suggest that the model has learned to decompose the grounding task into intermediate reasoning steps that are aligned with the linguistic constraints in the query. Fig. 8 serves as qualitative evidence that the proposed framework improves not only final grounding accuracy but also the interpretability of the model's decision-making process. This complements the quantitative results in Section IV as well as the training-log analysis in Figs. 6 and 7.

## VI. Discussion

Our results show that domain adaptation for RS VLMs benefits greatly from a two-stage approach: SFT with CoT data, followed by RL (GRPO). This method clearly outperformed both general-purpose models and single-stage baselines on in-domain and out-of-domain benchmarks.

Specifically, CoT-SFT gave the model explicit reasoning ability and domain knowledge, while the subsequent GRPO step further improved accuracy and robustness. The best results were consistently obtained by combining these stages (CoT-SFT+GRPO), with significant gains in both performance and interpretability. The ablation study also confirmed that diverse and balanced training data, as well as using reasoning-rich supervision, are key to better generalization.

These findings suggest that simply fine-tuning or applying RL to generic VLMs is insufficient for complex RS visual grounding tasks. Explicit reasoning supervision and curriculum-aware RL are both necessary.

For future work, expanding the quality and diversity of supervised CoT data, exploring richer reward signals, and applying this framework to more tasks such as open-vocabulary detection/segmentation and VQA are promising directions.

## VII. Conclusion

In this work, we proposed a two-stage training paradigm for RS VLMs, combining SFT with CoT data and RL. We also proposed approaches to generate high-quality CoT data for SFT and RL. Our approach effectively improves both the accuracy and interpretability of visual grounding in RS imagery. Experiments show that explicit reasoning supervision (CoT-SFT), together with RL (GRPO), is essential for robust generalization across different benchmarks. This framework provides a practical and scalable path for adapting general-purpose VLMs to specialized RS tasks, and lays the groundwork for future research in interpretable and reliable RSVLMs.

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